FMO Boundary Treatment and the Removal of Hybrid Orbitals by a Rectangular Transformation

Created: July 5, 2026

Purpose of this document

When a molecule is divided into fragments in the FMO method, a "forbidden hybrid orbital" pointing along a severed bond remains at the cut. This document describes how to remove that orbital from the calculation.

Stating the conclusion first: there is no need to push the orbital up with a large level shift B, as is done conventionally. Instead, one can solve the SCF from the outset in a smaller space that does not contain that direction. We call this operation "deflation by a rectangular transformation."

1 What is the problem in the first place?

FMO stands for Fragment Molecular Orbital. It is a method that performs quantum chemical calculations by dividing a large molecule into several small fragments. For systems such as proteins and polymers, which are too heavy to treat as a whole, this idea becomes essential.

However, when a molecule is divided into fragments, covalent bonds may be severed. For example, one may cut a carbon–carbon single bond in the middle, assigning the left side to fragment x and the right side to another fragment.

When a covalent bond is severed, the "direction that originally formed the bond" remains at the cut point. If the carbon is sp^3 hybridized, the problem is the one hybrid orbital, out of the four sp^3 hybrids, that points along the severed bond.

Schematically, the situation is as follows.

```
Fragment x side                  Other fragment side

   allowed directions -- C_alpha ---- [ cut-direction sp3 hybrid ]
                                                  ^
                                    not allowed in x calculation
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In this document we denote this "cut-direction hybrid orbital" by

$$\varphi_i^h.$$

If there are several of them, we number them $i = 1, 2, \ldots$

If this orbital is left usable in the calculation of fragment x, the electrons of fragment x leak into a direction that properly belongs to the partner fragment. This is physically undesirable.

Therefore, the FMO boundary treatment must do the following.

> Remove the hybrid orbitals pointing along the severed bonds from the SCF variational space of the fragment.

Here, the "variational space" is the set of degrees of freedom that the SCF is allowed to vary when searching for molecular orbitals. If an unwanted direction remains there, the SCF may end up using it.

2 A minimum of quantum chemistry vocabulary

2.1 AOs and MOs

In quantum chemical calculations, molecular orbitals (MOs) are expressed as linear combinations of atomic orbitals (AOs):

$$\psi_p = \sum_\mu C_{\mu p} \chi_\mu.$$

Here χ_μ are the AOs, ψ_p are the MOs, and $C_{\mu p}$ are the coefficients specifying how much of each AO is mixed into each MO.

The matrix collecting all MO coefficients is written $\mathbf{C}$. Roughly speaking, the SCF is a procedure that adjusts $\mathbf{C}$ to lower the electronic energy.

2.2 AOs are, in general, not orthogonal

The overlap between AOs is

$$S_{\mu\nu} = \langle \chi_\mu | \chi_\nu \rangle,$$

and the matrix collecting these is the overlap matrix $\mathbf{S}$.

If the AOs were perfectly orthogonal we would have $\mathbf{S} = \mathbf{I}$. In practice, however, with atom-centered basis sets, AOs on different atoms overlap in space. Hence, in general,

$$\mathbf{S} \neq \mathbf{I}.$$

This fact—that AOs are non-orthogonal—is the reason $\mathbf{S}$ appears repeatedly in the equations below.

2.3 The hybrid orbitals to be removed

The cut-direction hybrid orbitals can also be expressed as linear combinations of AOs:

$$\varphi_i^h = \sum_\mu A_{\mu i} \chi_\mu.$$

We write the matrix whose columns are these coefficients as $\mathbf{A}$. If N is the number of AOs and m is the number of hybrid orbitals to be removed, then $\mathbf{A}$ is an $N \times m$ matrix. For example, if there is only one cut-direction sp^3 hybrid, then $m = 1$ and $\mathbf{A}$ is a single column vector.

3 What does it mean to "not use the forbidden directions"?

Whether an MO ψ_p contains the hybrid orbital φ_i^h to be removed can be checked by their inner product:

$$\langle \varphi_i^h | \psi_p \rangle.$$

If this vanishes, ψ_p has no component along φ_i^h.

We want this to vanish for every removal target i and every MO p; in matrix form,

$$\mathbf{A}^\dagger \mathbf{S} \mathbf{C} = \mathbf{0}.$$

The reason $\mathbf{S}$ enters here is that the AOs are non-orthogonal. Indeed,

$$\begin{aligned}\langle\varphi_i^h|\psi_p\rangle &= \left\langle \sum_\mu A_{\mu i}\chi_\mu \middle| \sum_\nu C_{\nu p}\chi_\nu \right\rangle \\ &= \sum_{\mu\nu} A_{\mu i}^* \langle\chi_\mu|\chi_\nu\rangle C_{\nu p} \\ &= \sum_{\mu\nu} A_{\mu i}^* S_{\mu\nu} C_{\nu p}.\end{aligned}$$

Collecting this into matrix form gives $\mathbf{A}^\dagger\mathbf{SC}$.

The mathematical goal of the boundary treatment can therefore be summarized in a single line:

$$\boxed{\mathbf{A}^\dagger\mathbf{SC} = \mathbf{0}}$$

This states that "no MO contains a cut-direction hybrid orbital."

4 The conventional method: pushing up with a level shift

The conventional method assigns a very large energy to the forbidden orbitals. The idea is simple:

> Make using that direction so energetically costly that the SCF never chooses it.

This method is called level-shift projection, or hybrid orbital projection.

With the core Hamiltonian $\mathbf{H}$ and the environmental electrostatic potential $\mathbf{V}$, the level-shifted matrix takes the form

$$\widetilde{\mathbf{H}} = \mathbf{H} + \mathbf{V} + \mathbf{SABA}^\dagger\mathbf{S}.$$

Here $\mathbf{B}$ is a diagonal matrix of large positive values:

$$\mathbf{B} = \mathrm{diag}(B_1, B_2, \ldots, B_m).$$

Typically, very large values are used for B_i. This raises the orbital energies along the φ_i^h directions substantially, with the expectation that they are expelled from the occupied space.

4.1 Why the projection term has the form $\mathbf{SABA}^\dagger\mathbf{S}$

The component form of the projection term is

$$P_{\mu\nu} = \sum_i B_i \langle\chi_\mu|\varphi_i^h\rangle\langle\varphi_i^h|\chi_\nu\rangle.$$

The hybrid orbital φ_i^h is represented by the AO coefficients $\mathbf{A}$, but the AOs are non-orthogonal. Consequently,

$$\langle\chi_\mu|\varphi_i^h\rangle = (\mathbf{SA})_{\mu i},$$

and an $\mathbf{S}$ likewise emerges from the inner product on the right. Therefore

$$\mathbf{P} = \mathbf{SABA}^\dagger\mathbf{S}.$$

If the AOs were orthogonal from the start, with $\mathbf{S} = \mathbf{I}$, the projection term would simply be

$$\mathbf{ABA}^\dagger.$$

In other words, the two factors of $\mathbf{S}$ on the left and right are a consequence of using non-orthogonal AOs.

5 Weaknesses of the level-shift method

Increasing B_i makes the components of the unwanted hybrid orbitals smaller. With finite B_i, however, they never vanish completely. The residual components are roughly of size

$$O(B^{-1}).$$

On the other hand, making B_i too large places extremely large values inside the matrix. The condition number then deteriorates and the computation becomes numerically unstable. SCF convergence may also suffer.

Choice of B	**Advantage**	**Drawback**
Small	Numerically relatively stable	Unwanted components tend to remain
Large	Unwanted components become small	Condition number tends to deteriorate
Infinite	Ideally complete removal	Unusable in actual computation

To avoid this problem, a method that does not use B at all is needed.

6 The new idea: do not penalize—make the choice impossible from the start

The level-shift method "penalizes the forbidden direction whenever it is chosen." The rectangular-transformation approach changes the point of view:

Eliminate the forbidden directions from the variables from the very beginning.

The variables the SCF varies are the MO coefficients $\mathbf{C}$. So, instead of varying $\mathbf{C}$ directly and freely, we express it through a smaller coefficient matrix $\mathbf{C}_\mathrm{r}$:

$$\mathbf{C} = \mathbf{T}\mathbf{C}_\mathrm{r}.$$

Here $\mathbf{T}$ is a tall $N \times (N - m)$ matrix. Since the number of columns is $N - m$ rather than N, the number of degrees of freedom is reduced by m. This is why we call it a "rectangular transformation."

$\mathbf{T}$ is constructed so as to satisfy

$$\boxed{\mathbf{A}^\dagger\mathbf{S}\mathbf{T} = \mathbf{0}}$$

Then

$$\begin{aligned}\mathbf{A}^\dagger\mathbf{S}\mathbf{C} &= \mathbf{A}^\dagger\mathbf{S}\mathbf{T}\mathbf{C}_\mathrm{r} \\ &= \mathbf{0}\,\mathbf{C}_\mathrm{r} \\ &= \mathbf{0}.\end{aligned}$$

This is crucially important. Whatever value $\mathbf{C}_\mathrm{r}$ takes, the resulting $\mathbf{C} = \mathbf{T}\mathbf{C}_\mathrm{r}$ always satisfies

$$\mathbf{A}^\dagger\mathbf{S}\mathbf{C} = \mathbf{0}.$$

That is, no matter how the SCF proceeds, the unwanted hybrid orbitals are unusable from the start.

7 What does T represent?

T is the matrix whose columns are the allowed directions only. The condition characterizing the directions to be removed is

$$\mathbf{A}^\dagger \mathbf{S} \mathbf{v} = \mathbf{0}.$$

A vector **v** satisfying this is orthogonal, in the **S** metric, to the hybrid orbitals to be removed.

The space of all such **v** is written

$$\ker(\mathbf{A}^\dagger \mathbf{S}),$$

where ker stands for the kernel, i.e., the null space.

The columns of **T** form a basis spanning this kernel. In other words, **T** is "the matrix whose columns are exactly the directions orthogonal to the unwanted hybrid orbitals."

If the dimension of the full AO space is N and m hybrid orbitals are removed, the dimension of the remaining admissible space is

$$N - m.$$

Hence **T** is an $N \times (N - m)$ matrix.

8 An intuitive analogy

The difference between the two methods is easy to grasp through the following picture.

Method	**Analogy**	**What happens**
Level shift	Build a steep hill on the forbidden road	If the hill is finite, some intrusion remains possible
Rectangular transformation	Erase the road from the map	That direction simply cannot be taken

The level shift is a "penalty" method. The rectangular transformation is a "remove it from the options" method. Because of this difference, the unwanted components are zero in principle under the rectangular transformation.

9 With the rectangular transformation, the level-shift term vanishes

The level-shifted matrix was

$$\widetilde{\mathbf{H}} = \mathbf{H} + \mathbf{V} + \mathbf{SABA}^\dagger \mathbf{S}.$$

Let us transform it into the admissible space, i.e., multiply by $\mathbf{T}^\dagger$ from the left and **T** from the right:

$$\begin{aligned} \mathbf{T}^\dagger \widetilde{\mathbf{H}} \mathbf{T} = \mathbf{T}^\dagger (\mathbf{H} + \mathbf{V}) \mathbf{T} \\ + \mathbf{T}^\dagger \mathbf{SABA}^\dagger \mathbf{ST}. \end{aligned}$$

The second term can be rewritten as

$$\begin{aligned} &\mathbf{T}^\dagger \mathbf{SABA}^\dagger \mathbf{ST} \\ &\quad = (\mathbf{A}^\dagger \mathbf{ST})^\dagger \mathbf{B} (\mathbf{A}^\dagger \mathbf{ST}). \end{aligned}$$

But **T** is constructed so that

$$\mathbf{A}^\dagger \mathbf{ST} = \mathbf{0}.$$

Therefore

$$(\mathbf{A}^\dagger \mathbf{ST})^\dagger \mathbf{B} (\mathbf{A}^\dagger \mathbf{ST}) = \mathbf{0}.$$

Hence the matrix used in the reduced space is

$$\boxed{\mathbf{H}_\mathrm{r} = \mathbf{T}^\dagger(\mathbf{H}+\mathbf{V})\mathbf{T}}$$

and B no longer appears anywhere.

This is no accident. The role that B played—excluding the unwanted directions—has been taken over entirely by the condition on $\mathbf{T}$,

$$\mathbf{A}^\dagger\mathbf{S}\mathbf{T} = \mathbf{0}.$$

10 The SCF equations in the reduced space

The ordinary Roothaan–Hall equation is a generalized eigenvalue problem because the AOs are non-orthogonal. After the rectangular transformation, one solves the following equation in the reduced space:

$$\mathbf{F}_\mathrm{r}\mathbf{C}_\mathrm{r} = \mathbf{S}_\mathrm{r}\mathbf{C}_\mathrm{r}\boldsymbol{\varepsilon},$$

where

$$\mathbf{S}_\mathrm{r} = \mathbf{T}^\dagger\mathbf{S}\mathbf{T}$$

and

$$\mathbf{F}_\mathrm{r} = \mathbf{T}^\dagger\left[\mathbf{H}+\mathbf{V}+\mathbf{G}[\mathbf{D}]\right]\mathbf{T}.$$

$\mathbf{G}[\mathbf{D}]$ denotes the two-electron part, i.e., the Coulomb and exchange terms.

The density matrix is also mapped back from the reduced space to the original AO space before use:

$$\mathbf{D} = \mathbf{T}\mathbf{D}_\mathrm{r}\mathbf{T}^\dagger.$$

In this way, the two-electron integrals and the Fock-matrix construction are handled as quantities in the original AO space, while only the SCF degrees of freedom are reduced to dimension $N-m$.

11 S-orthonormalization

It suffices for the columns of $\mathbf{T}$ to be orthogonal, in the $\mathbf{S}$ sense, to the orbitals to be removed. As constructed, however, the overlap matrix of the reduced space,

$$\mathbf{S}_\mathrm{r} = \mathbf{T}^\dagger\mathbf{S}\mathbf{T},$$

need not be the identity.

One can therefore orthonormalize the columns of $\mathbf{T}$ in the $\mathbf{S}$ metric so that

$$\mathbf{T}^\dagger\mathbf{S}\mathbf{T} = \mathbf{I}.$$

In that case the SCF equation in the reduced space becomes an ordinary eigenvalue problem:

$$\mathbf{F}_\mathrm{r}\mathbf{C}_\mathrm{r} = \mathbf{C}_\mathrm{r}\boldsymbol{\varepsilon}.$$

In an implementation, one first finds a basis of

$$\ker(\mathbf{A}^\dagger\mathbf{S})$$

and then applies Löwdin or canonical orthogonalization to it to construct $\mathbf{T}$.

12 A concrete example: an sp^3 carbon

Simplify the valence AOs of carbon to just the four functions

$$s, \quad p_x, \quad p_y, \quad p_z.$$

From these four one can build four sp^3 hybrid orbitals.

In the ideal case they can be written, for example, as

$$\begin{aligned} h_1 &= \frac{1}{2}(s + p_x + p_y + p_z), \\ h_2 &= \frac{1}{2}(s + p_x - p_y - p_z), \\ h_3 &= \frac{1}{2}(s - p_x + p_y - p_z), \\ h_4 &= \frac{1}{2}(s - p_x - p_y + p_z). \end{aligned}$$

If the severed bond points along h_1, then h_1 must not be used in the calculation of fragment x. The allowed directions are then h_2, h_3, h_4. Locally, a dimensional reduction

$$4 \to 3$$

takes place.

In this simple orthogonal-basis example, the transformation matrix on the carbon is

$$\mathbf{T}_{\mathrm{C}} = \begin{array}{c|ccc} & h_2 & h_3 & h_4 \\ \hline s & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ p_x & \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} \\ p_y & -\frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ p_z & -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \end{array}$$

This $\mathbf{T}_{\mathrm{C}}$ can be viewed as the matrix obtained by dropping the h_1 column and keeping only the remaining three.

In actual calculations the basis set is much larger and the AOs are non-orthogonal, so the coefficients are not such clean values of 1/2. In practice one constructs $\mathbf{T}$ using $\mathbf{S}$ so that

$$\mathbf{A}^{\dagger}\mathbf{S}\mathbf{T} = \mathbf{0}.$$

Still, the idea is the same: "exclude the one cut-direction hybrid and build the MOs from the remaining allowed directions only."

13 Handling of nuclei and electron counts when a bond is severed

Both the rectangular transformation and the level-shift projection operate only on the electronic variational space, i.e., the MO coefficients $\mathbf{C}$. The nucleus of the α carbon itself is never divided by these operations. Since this point is easy to misunderstand, we summarize it in its own section.

13.1 Nuclei are not divided

Even when a covalent bond is severed at the α carbon, the carbon nucleus belongs, as a point charge, entirely to one of the fragments. Its nuclear charge is never halved, nor apportioned according to the number of hybrid orbitals. Of the four hybrids, only one (h_1) is removed; this merely reduces by one the number of directions the electrons can use, and neither the position of the nucleus nor the nuclear charge Z_{C} changes.

```
nucleus  Z_C = +6   <- point charge; belongs wholly to fragment x
   |                    (never divided)
s, px, py, pz       <- all four AOs remain on the carbon
   |
only the single direction h1 is removed
(neither the nucleus nor the AOs themselves)
```

If the α carbon is assigned to fragment x, the pieces are distributed as follows.

- The nuclear charge $+Z_{\mathrm{C}}$ enters the core Hamiltonian $\mathbf{H}$ of fragment x (as nucleus–electron attraction).
- The nuclei on the partner-fragment side enter the environmental electrostatic potential $\mathbf{V}$.
- The AOs on the carbon (s, p_x, p_y, p_z and their generalizations) all remain in the basis. What the rectangular transformation removes is only "one direction" within the space these AOs span.

In short, neither nuclei nor AOs are divided or deleted; what is transformed is only the set of directions available to the electrons.

13.2 The number of AOs does not decrease; what decreases is the number of variational degrees of freedom

This point is particularly prone to confusion, so we emphasize it. The rectangular transformation does not physically reduce the number of AO basis functions N. The basis functions remain N in number, and the two-electron integrals are computed in the full N-dimensional space. What is reduced is the number of degrees of freedom the SCF may vary in searching for the MOs: from N to $N-m$. In terms of the equations, it is simply that $\mathbf{C}_{\mathrm{r}}$ in $\mathbf{C} = \mathbf{T}\mathbf{C}_{\mathrm{r}}$ is $(N-m)$-dimensional.

13.3 Which fragment keeps the two electrons of the severed bond?

When one covalent bond is severed, the assignment of the electron pair (two electrons) that occupied that bond must be decided separately. Removing a hybrid orbital only removes a "direction"; it does not by itself determine the electron count. In the standard FMO procedure, the electron pair belonging to the severed bond is assigned to one of the fragments, and the treatment of the boundary carbon (the so-called bond-detached atom) is made consistent with that choice.

An implementation must therefore keep the following three items mutually consistent:

1. To which fragment the nucleus (and core electrons) of the α carbon are assigned.
2. To which fragment the electron pair of the severed bond is assigned.
3. Which hybrid orbitals are removed from the variational space.

The rectangular transformation $\mathbf{T}$ (or the level shift $\mathbf{B}$) is responsible only for the third item. The first and second are fixed separately as part of the definition of the fragment partitioning.

14 Similarities to and differences from the $6 \to 5$ transformation of d functions

In quantum chemistry one sometimes transforms the six Cartesian d functions into the five spherical-harmonic d functions. This is also a $6 \to 5$ dimensional reduction, so formally it resembles the rectangular transformation. The meaning, however, is different.

Aspect	**Cartesian d to spherical d**	**FMO boundary rectangular transformation**
Dimensions	$6 \to 5$	$N \to N - m$
What is dropped	The redundant s-like component mixed into the d functions	The hybrid orbital along the severed bond
How the matrix is determined	An essentially universal transformation fixed by angular momentum	Depends on the bond direction, **A**, and **S**
Purpose	Tidy up the representation of the basis functions	Remove unwanted components at the fragment boundary

Both reduce dimensions with a rectangular matrix, and in that respect they are alike. However, the **T** of the FMO boundary treatment is a matrix that imposes a physical constraint determined system by system.

15 Relation to the level-shift method

The rectangular-transformation approach is not an unrelated alternative to the level-shift method. Rather, it can be understood as follows:

$$\boxed{\text{deflation} \iff B \to \infty}$$

That is, the rectangular transformation corresponds to the $B \to \infty$ limit of the level-shift method.

In the level-shift method, the larger B is, the smaller the components of the unwanted orbitals become. Ideally, taking B to infinity would remove them completely. But an infinite B cannot be used in actual numerical computation, and using a very large finite value worsens the condition number.

The rectangular transformation realizes this "ideal state of infinite B" without actually using B. As a result, the tuning problem of "which value to choose for the level shift B" disappears altogether.

16 Where do the matrices become non-square?

A natural question at this point is: "In the original approach with the level shift B, do the row and column dimensions also change, making the matrices non-square?" The short answer is:

> In the level-shift method every matrix remains square, $N \times N$. Rectangular matrices appear only in the rectangular-transformation approach, and even there the rectangular one is the transformation matrix **T** itself; the matrix that is finally diagonalized is again square.

16.1 Level-shift method: the shape stays $N \times N$

The level-shifted matrix

$$\widetilde{\mathbf{H}} = \mathbf{H} + \mathbf{V} + \mathbf{SABA}^{\dagger}\mathbf{S}$$

is an $N \times N$ square matrix. Tracking the sizes of the projection term $\mathbf{SABA}^{\dagger}\mathbf{S}$,

$$\underbrace{\mathbf{S}}_{N\times N}\,\underbrace{\mathbf{A}}_{N\times m}\,\underbrace{\mathbf{B}}_{m\times m}\,\underbrace{\mathbf{A}^{\dagger}}_{m\times N}\,\underbrace{\mathbf{S}}_{N\times N} \longrightarrow N \times N.$$

Since m appears in the middle, the rank is m, but the "shape"—rows $\times$ columns—remains $N \times N$.

In the level-shift method, therefore:

- The generalized eigenvalue problem to be solved stays $N \times N$; the dimension is not reduced.
- N MOs are still obtained. The unwanted hybrid-orbital directions remain among the N solutions as high-lying virtual orbitals whose energies have been raised by B. They do not disappear with a reduction of dimension; they are merely "pushed up and left unoccupied."

16.2 Rectangular transformation: only T is rectangular

In the rectangular-transformation approach, the transformation matrix

$$\mathbf{T}: \quad N \times (N-m) \qquad \text{(rectangular)}$$

is the only rectangular matrix. Sandwiching with it, the transformed matrices

$$\mathbf{F}_{\mathrm{r}} = \mathbf{T}^{\dagger}\mathbf{F}\mathbf{T}: \ (N-m) \times (N-m), \qquad \mathbf{S}_{\mathrm{r}} = \mathbf{T}^{\dagger}\mathbf{S}\mathbf{T}: \ (N-m) \times (N-m)$$

are again square, though their size has shrunk from N to $N-m$.

In other words, a "rectangular matrix with different row and column dimensions" appears only as the $\mathbf{T}$ of the rectangular transformation; the Fock and overlap matrices actually diagonalized are square in both methods.

Matrix	**Level-shift projection**	**Deflation by rectangular transformation**
Transformation matrix	None (effectively the identity $\mathbf{I}$, $N \times N$)	$\mathbf{T}$: $N \times (N-m)$ (rectangular)
Matrix diagonalized	$\widetilde{\mathbf{H}}$: $N \times N$ (square)	$\mathbf{F}_{\mathrm{r}}, \mathbf{S}_{\mathrm{r}}$: $(N-m) \times (N-m)$ (square)
Number of MOs	N	$N-m$
Treatment of unwanted directions	Remain among the N solutions at high energy	Simply absent from the space

16.3 Another viewpoint: the column count of T is the answer

That $\mathbf{T}$ is rectangular is itself the expression of the fact that m directions have been dropped. A square $N \times N$ matrix is a transformation that "keeps all N directions," and the level shift corresponds to exactly this (the identity transformation plus a projection term). A rectangular $N \times (N-m)$ matrix is a transformation that "keeps only $N-m$ directions," and this is the essence of deflation.

The correspondence of the previous section,

$$\text{deflation} \iff B \to \infty,$$

can be restated in these terms as well. In the $B \to \infty$ limit, m directions of the level shift's $N \times N$ space are raised infinitely high and become effectively unusable, and the remaining $N-m$ dimensions coincide exactly with the reduced space of the rectangular transformation.

17 Comparison of the two methods

Item	**Level-shift projection**	**Deflation by rectangular transformation**
Basic idea	Penalize unwanted directions energetically	Remove unwanted directions from the variables
Condition used	Add $\mathbf{SABA}^\dagger\mathbf{S}$ to the Fock matrix	Use a $\mathbf{T}$ satisfying $\mathbf{A}^\dagger\mathbf{ST} = \mathbf{0}$
B	Required	Not required
Exactness of removal	Approximate for finite B	Exact in principle
Residual components	Roughly $O(B^{-1})$	Zero to machine precision
Dimension	Stays N	$N - m$
Numerical stability	Condition number can deteriorate for large B	No B-induced conditioning problems
Viewpoint	Push out by making B large	Realize the $B \to \infty$ limit with a finite transformation

18 Implementation outline

An implementation proceeds as follows.

1. Build the cut-direction hybrid orbitals as AO coefficients: $\mathbf{A}$.
2. Prepare the AO overlap matrix: $\mathbf{S}$.
3. Find a basis spanning the kernel: $\ker(\mathbf{A}^\dagger\mathbf{S})$.
4. If desired, $\mathbf{S}$-orthonormalize that basis to obtain a $\mathbf{T}$ satisfying $\mathbf{T}^\dagger\mathbf{ST} = \mathbf{I}$.
5. Transform the Fock and overlap matrices into the reduced space:
$$\mathbf{F}_\mathrm{r} = \mathbf{T}^\dagger\mathbf{FT}, \qquad \mathbf{S}_\mathrm{r} = \mathbf{T}^\dagger\mathbf{ST}.$$
6. Solve the SCF equation in the reduced space:
$$\mathbf{F}_\mathrm{r}\mathbf{C}_\mathrm{r} = \mathbf{S}_\mathrm{r}\mathbf{C}_\mathrm{r}\boldsymbol{\varepsilon}.$$
7. Map back to the original AO space as needed:
$$\mathbf{C} = \mathbf{TC}_\mathrm{r}, \qquad \mathbf{D} = \mathbf{TD}_\mathrm{r}\mathbf{T}^\dagger.$$

Nowhere in this procedure does the level shift B appear.

19 Summary

In FMO, fragments are created by severing covalent bonds, so the hybrid orbitals pointing along the severed bonds must be removed from the SCF of the corresponding fragment.

The conventional level-shift method assigns a large energy B to the unwanted hybrid orbitals to expel them from the occupied space. With finite B, however, the unwanted components never vanish completely, and making B too large causes numerical instability.

In the rectangular-transformation approach, one constructs a matrix $\mathbf{T}$ that spans only the space $\mathbf{S}$-orthogonal to the unwanted hybrid orbitals, and sets

$$\mathbf{C} = \mathbf{TC}_\mathrm{r}.$$

Since

$$\mathbf{A}^\dagger \mathbf{S} \mathbf{T} = \mathbf{0},$$

we have, for any $\mathbf{C}_\mathrm{r}$,

$$\mathbf{A}^\dagger \mathbf{S} \mathbf{C} = \mathbf{0}.$$

Therefore the unwanted hybrid orbitals never mix into the MOs in the first place.

This method realizes the $B \to \infty$ limit of the level-shift method with finite matrix operations and without using B. As a result, the need to tune the level-shift value disappears, and the trade-off between residual components and conditioning problems is avoided.

List of symbols

Symbol	Meaning
$\mathbf{H}$	Core Hamiltonian
$\mathbf{V}$	Environmental electrostatic potential from the other fragments, etc.
Z_C	Nuclear charge of the boundary (α) carbon; not divided, belongs to one fragment
$\mathbf{S}$	AO overlap matrix
χ_μ	AO basis function
ψ_p	MO
$\mathbf{C}$	MO coefficients in the original AO space
$\mathbf{C}_\mathrm{r}$	MO coefficients in the reduced space
φ_i^h	Hybrid orbital to be removed
$\mathbf{A}$	AO coefficients of the hybrid orbitals to be removed
m	Number of hybrid orbitals to be removed
N	Number of AO basis functions
$\mathbf{B}$	Level-shift matrix
$\mathbf{T}$	Rectangular transformation matrix spanning the admissible space only
$\mathbf{F}$	Fock matrix
$\mathbf{G}[\mathbf{D}]$	Two-electron term
$\mathbf{D}$	Density matrix
$\boldsymbol{\varepsilon}$	Orbital energies